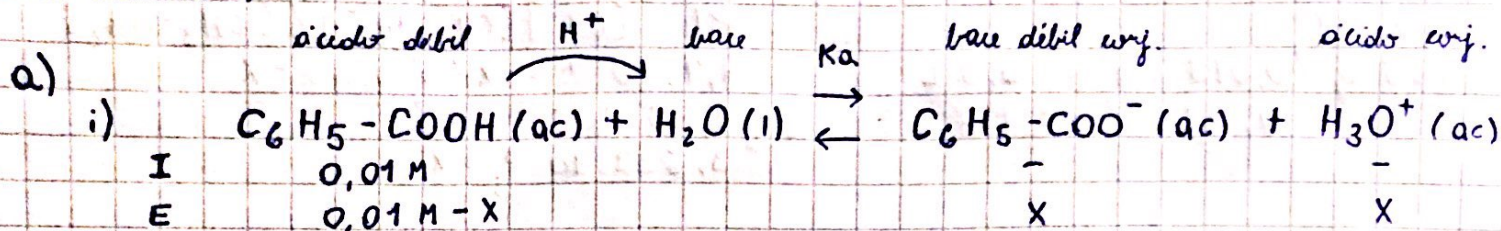


PROBLEMA 4



$$K_a = \frac{[C_6H_5-COO^-][H_3O^+]}{[C_6H_5-COOH]}$$

$$K_a = \frac{x^2}{0,01 M - x}$$

Disociado al 7,75% $\rightarrow \frac{x}{0,01} = 0,0775 = \alpha$

$$\alpha = x / [C_6H_5-COOH]_0$$

$$x = \alpha \cdot [C_6H_5-COOH]_0$$

$$x = 0,0775 \cdot 0,01 M$$

$$x = 7,75 \cdot 10^{-4} M$$

$$pH = -\log [H_3O^+]$$

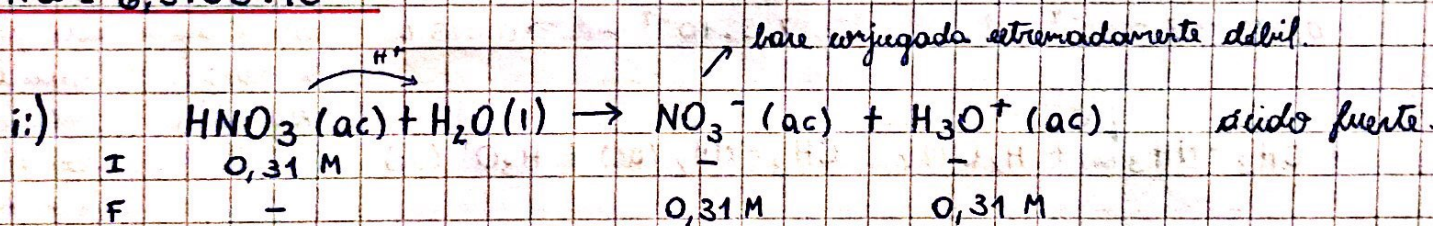
$$pH = -\log x$$

$$pH = 3,11$$

$$K_a = \frac{(7,75 \cdot 10^{-4})^2}{7,75 \cdot 10^{-4} + 0,01}$$

$$K_a = \frac{6,0625 \cdot 10^{-7}}{9,225 \cdot 10^{-3}}$$

$$K_a = 6,5108 \cdot 10^{-5}$$



$$pH = -\log 0,31$$

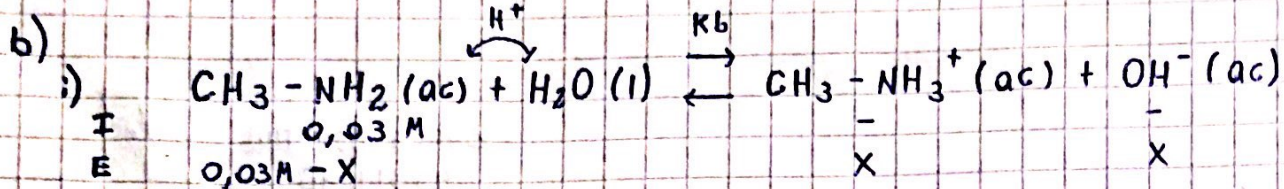
$$pH = 0,5 < 3,11$$

$$0,31 / 7,75 \cdot 10^{-4}$$

Para que el ácido nítrico alcance el mismo pH que el ácido benzoico su concentración debe disminuir 400 veces. De esa forma igualar las concentraciones y por tanto pH.

$$\begin{array}{l} \text{ácido benzoico} = 7,75 \cdot 10^{-4} \text{ moles / 1 L} \\ \text{ácido nítrico} = 0,31 \text{ moles / 1 L} \end{array} \quad \left. \begin{array}{l} \nearrow \\ \searrow \end{array} \right\} 400 <$$

$$[HNO_3] \gg [C_6H_5-COOH]$$

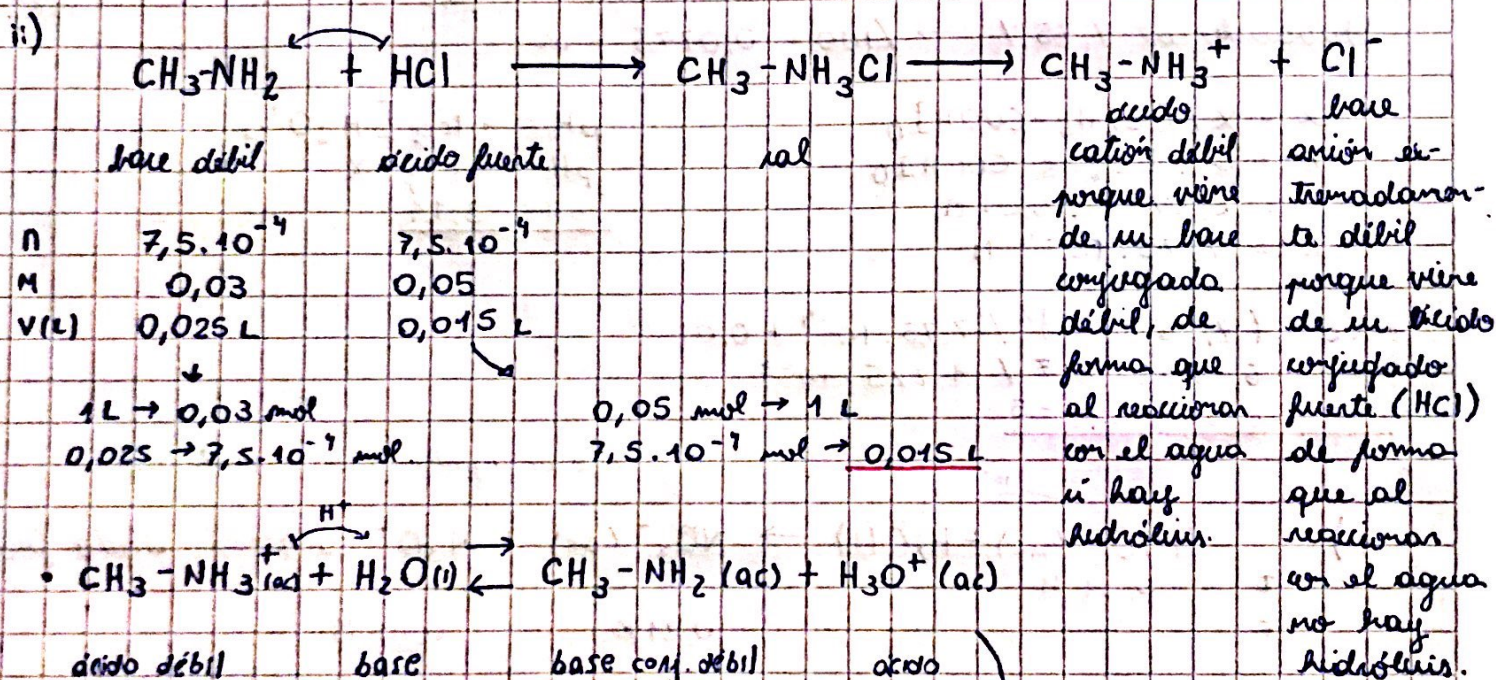


31 g/mol
 $0,465 \text{ g} / 0,015 \text{ mol} \leftarrow 0,5 \text{ L}$
 $0,03 \text{ mol} \leftarrow 1 \text{ L}$

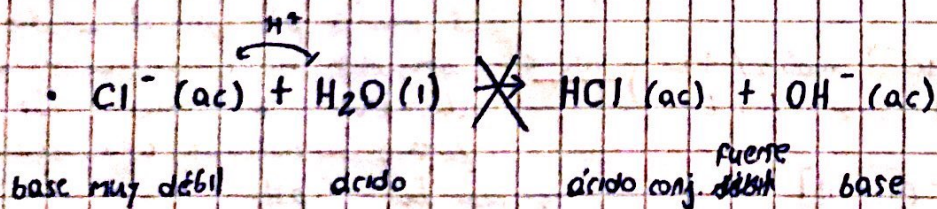
$K_b = [\text{OH}^-][\text{CH}_3\text{-NH}_3^+] / [\text{CH}_3\text{-NH}_2]$
 $4,4 \cdot 10^{-4} = X^2 / 0,03 - X$
 $\sqrt{1,32 \cdot 10^{-5} = X^2}$
 $3,633 \cdot 10^{-3} = X$

$\text{pOH} = -\log[\text{OH}^-]$
 $\text{pOH} = -\log 3,633 \cdot 10^{-3}$
 $\text{pOH} = 2,439$

$14 - \text{pOH} = \text{pH}$
 $14 - 2,439 = \underline{11,56} = \text{pH}$

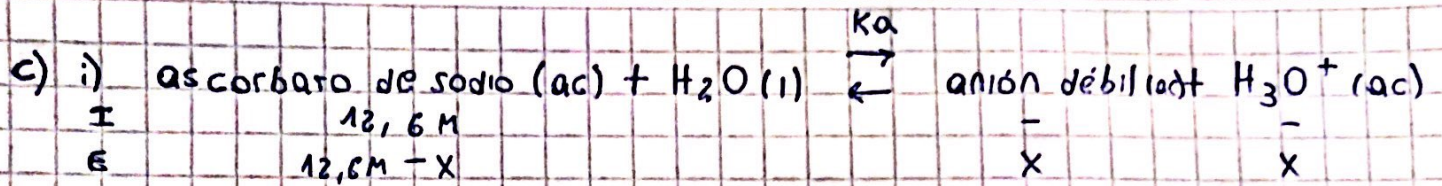


$[\text{H}_3\text{O}^+] > [\text{OH}^-]$
 $\text{pH} < 7$
Sc. ácida



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[Signature]



198 g / mol.

500 g / 2,52 mol $\rightarrow$ 0,2 L
 12,6 mol $\leftarrow$ 1 L

$pK_a = 4,1$

$K_a = 10^{-4,1} = 7,943 \cdot 10^{-5}$

$$K_a = [\text{anión}][\text{H}_3\text{O}^+] / [\text{ascorbato de sodio}]$$

$$7,943 \cdot 10^{-5} = X^2 / 12,6 - X$$

$$1,10^{-3} = X^2$$

$$0,031 = X$$

$$pH = -\log [\text{H}_3\text{O}^+]$$

$$pH = -\log 0,031$$

$$pH = \underline{1,49}$$

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